

CStats - Lab #7

April 21, 2026

Exercise 1.

The MLE of σ^2 in simple linear regression is $\widehat{\sigma^2} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$, but we instead use

$$\text{MSE} = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

as an unbiased estimator of σ^2 . Intuitively, why do we divide by $n-2$ rather than n ? Why specifically the number 2?

Exercise 2.

The LINE assumptions for linear regression are Linearity, Independence, Normality, and Equal variance. The normality assumption is often stated as “the errors ε_i are normally distributed.” Why is this assumption needed, and at what stage of the analysis does it become essential? Is it needed to *compute* the least-squares estimates?

Exercise 3.

A regression model reports $R^2 = 37.04\%$. What does this number mean intuitively? Does a high R^2 guarantee that the model is “good”? Does a low R^2 mean the model is useless?

Exercise 4.

Deep learning is presented as a three-step procedure: (1) define a set of functions, (2) assess the goodness of each function, (3) pick the best function. Map these three steps onto ordinary linear regression. What plays the role of the “function set,” the “loss,” and the “picking” step in linear regression?

Exercise 5.

For linear regression we derived closed-form formulas for a and b by setting derivatives to zero. Why, then, do deep-learning courses spend so much time on gradient descent? Give at least two reasons why the closed-form trick does not extend to neural networks.

Exercise 6.

The lecture poses the following claim: “Every time we update the parameters by gradient descent we obtain θ that makes $L(\theta)$ smaller, so $L(\theta^0) > L(\theta^1) > L(\theta^2) > \dots$.” Is this statement correct? If not, under what conditions can it fail?

Exercise 7.

In the Pokémon CP example, polynomial models of degree 1 through 5 gave these errors:

degree	Training err.	Testing err.
1	31.9	35.0
2	15.4	18.4
3	15.3	18.1
4	14.9	28.2
5	12.8	232.1

Explain, intuitively, why training error keeps dropping while testing error eventually explodes. Which model would you deploy, and why?

Exercise 8.

Regularization adds a penalty to the loss:

$$L = \sum_n \left(\hat{y}^n - (b + \sum_i w_i x_i) \right)^2 + \lambda \sum_i w_i^2.$$

- (a) Why does this encourage “smoother” functions? (b) Should the bias b also be penalized?
(c) How should one pick λ ?

Exercise 9.

Draw the simplest possible neural network that implements ordinary linear regression $\hat{y} = b + w x$. How many layers does it have? What is the activation function? What does this tell you about the relationship between linear regression and deep learning?